

From: Brady, Donal (DEQ)
Sent: 2/5/2019 4:43:37 PM
To: ""Doug Langworthy"" <dlangworthy@Cadillac-MI.net>; "Cindy Tomaszewski" <ctomaszewski@Cadillac-MI.net>
Cc: "Jeff Dietlin" <utilities@Cadillac-MI.net>
Subject: RE: City of Cadillac WWTP 2018 DEQ Inspections
Attachments: Cadillac WWTP 2018 Inspection Report Final.docx, 2018 Cadillac WWTP CSI Final Report.xlsx, Basis for Decision Memo-DRAFT_Cadillac WWTP.pdf, Cadillac Asset Mngmnt Submittals 2016 2017 2018.pdf

Hi Doug,

Thanks for this info. and also for meeting me at the plant on 12/14/18. The purpose of this email is to provide the City with information on the 2018 inspection, an update on the permit renewal, and information of past NPDES permit violations. Please review this email and respond according to the list at the bottom of the email.

2018 INSPECTION

The revised and final inspection form and sampling report are attached. The City is doing a great job at the WWTP.

NPDES PERMIT RENEWAL

With respect to the WWTP's NPDES review, the No Exposure Certification (NEC) form for stormwater should be modified. Item 9 of the NEC form submitted in MiWaters (form submission number "HNH-YJTD-MY6A8" and form status "In Process") should be changed from "Yes" to "No." I understand from our meeting on 12/14/18 that Item 9 was answered "Yes" due to the liquid/leachate offloading that occurs outdoors at the plant. However, during this loading, materials or pollutants do not normally contact stormwater, would only contact stormwater during a spill, and even if stormwater contacted a spill it is very unlikely the stormwater would flow to a storm sewer or surface water body.

Also, DEQ permit staff are working on the draft version of a new permit that could include four seasons or the current three seasons. The four seasons proposed by permit staff and preliminary limits for CBOD and ammonia for those seasons are shown in the table below and in the attached memo. from DEQ-WRD permit staff. Also note, permit staff is proposing limits for copper, nickel, selenium, and bis(2-ethylhexyl)phthalate in the new permit because plant data indicates the effluent could possibly exceed standards for these chemicals.

Season	CBOD Maximum Limits for Loading [lbs/day]		CBOD Maximum Limits for Concentration [mg/L]		Ammonia Maximum Limits for Loading [lbs/day]		Ammonia Maximum Limits for Concentration [mg/L]	
	Monthly	7-Day	Monthly	7-Day	Monthly	7-Day	Monthly	7-Day
June-September	190	270	7	10	24	53	0.9	2.0

October-November	290	480	11	17	88	--	1.6	2.8
December-March	670	910	25	34	99	---	3.7	---
April-May	450	670	17	25	130	---	4.8	---

PAST NPDES PERMIT VIOLATIONS

The DEQ is trying to cleanup some internal files in MiWaters. The DEQ needs to show communications were sent regarding past permit violations (looking back approximately to 2017). Please see the following table of the violations noted in MiWaters for 2017 and later (violations related to biosolids and the industrial pretreatment program are not listed). Violations that still need to be addressed are highlighted in yellow.

Facility Name and Permit Number – Cadillac WWTP MI0020257			
<i>The Michigan Department of Environmental Quality (DEQ), Water Resources Division (WRD), recently conducted a MiWaters file review of the compliance history for the facility identified above. Based on this review, the following violations were noted in 2017, 2018, and 2019 (to date). Please note, violations related to biosolids and the industrial pretreatment program are not listed below.</i>			
<i>Note, violations below that are shaded yellow require a formal response.</i>			
Violation Description	Date	Summary	Significant Non-compliance
CBOD exceedance (due to lab error), CBOD 10.58 mg/L, limit was 10 mg/L	11/19/18	Written report received 11/26/18, violation reviewed and determined to be an error, result likely not representative of discharge, no further action required unless problem reoccurs	Inactive – error
Annual asset management report submitted a day late	7/30/18	Violation being reviewed, additional asset management information is required	No
CBOD exceedance related to cold weather, CBOD 11.40 mg/L, limit was 10 mg/L	5/4/18	Written reports received 5/7-9/18, violations reviewed, additional seasons will be considered for new permit in an effort to address late spring, early	No
CBOD exceedance related to cold weather, CBOD 12.60 mg/L, limit was 10 mg/L	5/3/18		
CBOD exceedance related to cold weather, CBOD 11.60 mg/L, limit was 10 mg/L	5/2/18		

		winter, etc., no further action required unless exceedances reoccur after seasons added to permit	
CBOD exceedance due to sample pump malfunction, CBOD 11.10 mg/L, limit was 10 mg/L	11/27/17	Written report received 12/4/17, violation reviewed and determined to be an error, results likely not representative of discharge, additional pump purging implemented to prevent reoccurrence, no further action required unless problem continues to reoccur	Inactive – error
CBOD exceedance related to filamentous bacteria, CBOD 22.50 mg/L, limit was 10 mg/L CBOD exceedance related to filamentous bacteria, CBOD 11.30 mg/L, limit was 10 mg/L	10/24/17 10/20/17	Written reports received 10/25&30/17, violations reviewed, WWTP applied low level chlorine dose, adjusted RAS, adjusted WAS, and utilized EQ to control filamentous, no further action required unless problem continues to reoccur	No
TSS 7-Day Average 1929.67 lbs/day, limit was 800 lbs/day CBOD 7-Day Average 512.8 lbs/day, limit was 270 lbs/day TSS 7-Day Average 84.67 mg/L, limit was 30 mg/L TSS 7-Day Average 33.33 mg/L, limit was 30 mg/L	10/24/17 10/20/17	Violations being re-evaluated, Doug and Cindy, I think these are data entry (calculation) errors, we should try and correct the October 2017 DMR. Also, I looked at the December 2018	No

		DMR and have some questions about the calculations.	
CBOD 7-Day Average 276.7 lbs/day, limit 270 lbs/day	7/6/17	Violation being re-evaluated, Doug and Cindy, I think this is a data entry error, we should try and correct the July 2017 DMR.	No
CBOD exceedance due to bad nutrient water used in analysis, blank showed high depletion, CBOD 13.90 mg/L, limit was 10 mg/L	7/6/17	Written report received 7/12/17, violation reviewed and determined to be an error, plant changed source of nutrient water, no further action required unless problem continues to reoccur	Inactive - error
CBOD exceedance due to bad nutrient water used in analysis, blank showed high depletion, CBOD 14.00 mg/L, limit was 10 mg/L	7/3/17		
CBOD exceedance, related to sample refrigerator malfunction, CBOD 13.00 mg/L, limit was 10 mg/L	6/9/17	Written report received 6/15/17, violation reviewed and determined to be an error, plant changed source of nutrient water, no further action required unless problem continues to reoccur	Inactive - error
CBOD exceedances related to cold weather, limit was 10 mg/L		Written reports received 5/12 & 24/17, violations reviewed, additional seasons will be considered for new permit in an effort to address late spring, early winter, etc., no further action required unless exceedances reoccur after seasons added to permit	No
CBOD 11.70 mg/L	5/19/17		
CBOD >17.400 mg/L (this result elevated due to sample pump malfunction)	5/18/17		
CBOD 11.70 mg/L	5/17/17		
CBOD 10.20 mg/L	5/15/17		
CBOD 13.80 mg/L	5/5/17		
CBOD 10.70 mg/L	5/4/17		

CBOD 11.20 mg/L	5/3/17		
CBOD Monthly Average 8.95 mg/L, limit was 7 mg/L; Value: 8.95	May 2017		
CBOD 7-Day Average >341.626 lbs/day, limit was 270 lbs/day	5/18/17	Violation being re-evaluated, Doug and Cindy, I think this is a data entry error, we should try and correct the May 2017 DMR.	No
Ammonia exceedance related to sample pump malfunction, Ammonia 4.62 mg/L, limit was 2 mg/L	5/18/17	Written report received 5/24/17, violation reviewed and determined to be an error, results likely not representative of discharge, no further action required unless problem continues to reoccur	Inactive - error

RESPONSE REQUIRED FROM THE CITY OF CADILLAC

1. Please modify Item 9 on the version of the stormwater NEC form with the submission number HNH-YJTD-MY6A8 (please note there are two versions of this form is MiWaters)
2. Review the proposed permit limits on the attached memo. from DEQ permit staff. Please let me know if you have questions or comments, also let me know if you are interested in the permit incorporating three or four seasons. Please get back to me on the permit limits by 2/15/19. Note the City also have a time period to comment once the draft permit is published by permit staff.
3. Provide the DEQ with additional asset management information listed below.
 - a. An asset management plan meeting the requirements of Part I, Section A.5.a. of the NPDES permit MI0020257. The 2016 asset management report submitted by the City stated an asset management plan would be written in the summer of 2016 but no plan was submitted to the DEQ.
 - b. An up-to-date collection system map or written confirmation the collection system map as stated in Part I, Section A.5.a.2 of the NPDES permit MI0020257.
 - c. An inventory and assessment of fixed assets, including the information required in Part I, Section A.5.a.3 of the NPDES permit.
 - d. An OM&R budget and rate sufficiency demonstration, including the information required in Part I, Section A.5.a.4 of the NPDES permit.
4. Check the DMR calculations for the May 2017, July 2017, and October 2017 permit limit exceedances highlighted yellow above.
5. Check the DMR calculations for December 2018 and the calculations being used for future DMRs.

Please let me know what questions you have about this email. I am also available to meet with you to discuss this. DEQ-WRD permit staff in Lansing is working on the draft NPDES permit, please let me know your thoughts on the proposed permit limits and associated seasons by the end of next week. Also, the City has provided limited asset management information to the DEQ (a total of 5 pages, see attached). Therefore, asset management requirements will be maintained in the renewed NPDES permit. I know the City is implementing an asset management program and it would be unfortunate if we had to write more formal letters because some details have not been provided to the DEQ. Please send me or show me the asset management information required by the NPDES permit and requested above.



Don Brady, P.E.
Michigan Department of Environmental Quality
Water Resources Division
Cadillac District Office
120 West Chapin Street
Cadillac, MI 49601
BradyD6@michigan.gov
(231) 876-4478

From: Doug Langworthy <dlangworthy@Cadillac-MI.net>
Sent: Wednesday, October 24, 2018 10:59 AM
To: Brady, Donal (DEQ) <BradyD6@michigan.gov>; Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>
Cc: Jeff Dietlin <utilities@Cadillac-MI.net>
Subject: RE: City of Cadillac WWTP 2018 DEQ Inspections

Don,

Sorry it took as long as it did but I think we have all the answers to your questions.

- Item 5 on page 8 – See attached PDF
- Item 13 on page 9 - Ok
- Item 15 on page 12 – See attached PDF
- Item 32 on page 14 – See attached PDF
- Item 11 on page 1 – Yes, only two chemicals you have listed ferric 38.5% and Sodium hypochlorite 90% tablet.
- Item 8 on page 16 – See attached spreadsheet
- Item 1 on page 17 – 60% Residential and 40% Industrial
- Items 2h, 2i, 2j, and 2o on page 23 – 2h Yes, flows freely
 - 2i Yes, the channel is 4 times longer than the measurable level in the channel.
 - 2j Yes, the distance is 2 times the measurable level in

the channel.

2o Q (inCFS)=3.367 LH 3/2 See attached photo

The attached results for the composite BOD do vary but look ok.

If you need anything else please feel free to contact me. I hope this helps.

Doug

Douglas Langworthy

Assistant Water Resources Supervisor

City of Cadillac

231-775-2841

From: Brady, Donal (DEQ) [<mailto:BradyD6@michigan.gov>]

Sent: Thursday, October 11, 2018 5:23 PM

To: Doug Langworthy; Cindy Tomaszewski

Cc: Jeff Dietlin

Subject: City of Cadillac WWTP 2018 DEQ Inspections

Hi Doug and Cindy,

Thanks for taking the time to review the WWTP with me earlier this year. The plant looked great and you (and your staff) are doing a great job. Please see the draft inspection form. Could you please look this over and let me know if you see anything that needs to be corrected? Could you pay particular attention to the following?

- Item 5 on page 8
- Item 13 on page 9
- Item 15 on page 12
- Item 32 on page 14

Could you please also provide the information for the following?

- Item 11 on page 1
- Item 8 on page 16
- Item 1 on page 17
- Items 2h, 2i, 2j, and 2o on page 23

(Don't worry about the pump station information on pages 18 and 19)

Also, please find attached the results of the sampling/analysis completed by the DEQ. Could you please let me know what you think of these results (especially the composite BOD results)?

Finally, in order to cross an administrative T (dot an administrative I) we should fill out a No Exposure Certification (NEC) form for stormwater. Stormwater from the Cadillac plant is regulated because the design flow is > 1 MGD. However, if stormwater that falls on the plant does not contact polluting materials we can fill out a NEC vs. including the stormwater management and monitoring requirements in the plant's NPDES permit. Could you please submit a NEC form in MiWaters? You should be able to find the form under "Apps, Requests and Reports" and then "All other forms." If you think the plant has already completed a NEC, please let me know.

Please let me know if you have any questions. Thanks for working with me on this inspection.



Don Brady, P.E.
Michigan Department of Environmental Quality
Water Resources Division
Cadillac District Office
120 West Chapin Street
Cadillac, MI 49601
BradyD6@michigan.gov
(231) 876-4478

ATTACHMENT NAME:

Cadillac WWTP 2018 Inspection Report Final.docx

ATTACHMENT TYPE:

Microsoft Office 12+, Office Open XML (OOXML/OpenXML): document (DOCX)

Cadillac WWTP : 15637

NPDES - Facility Site Review

Inspector: Donal Brady

Start Date: 05/17/2018 Completed Date: 09/28/2018

Inspection Information

1. Permit Number: MI0020257

2. Date: 05/17/2018

3. Time: 9:00 AM

4. Inspection Participants Present

Name	Title	Affiliation	Phone Number
Doug Langworthy	Assistant Water Resources Supervisor	City of Cadillac	2317752841
Cindy Tomaszewski	Laboratory Supervisor	City of Cadillac	2317752368

5. Opening Conference Notes:

I explained basic inspection procedure. Doug and Cindy are experienced/knowledgeable personnel familiar with the DEQ. I reviewed the inspection forms/plant with Doug first then Cindy.

6. Facility classification (classification document on file)

Class A, treatment facilities serving or designed to serve a population in excess of 50,000 persons.

Per MiWaters site details page, the plant is Class B by population served but was previously designated Class A due to the plant complexity and the discharge to the Clam River (trout stream)

7. Designated Certified Operator

Certification Type	Number	Exp. Date	Reported to DEQ?
A, B, C, D	10435	04/15/2020	X
C, D	18398	07/15/2019	X

Operator is Jeff Dietlin (#10435), Doug's certification # is #18398. The City sent the DEQ a list of operators in Feb-2016.

The facility has provided written notification for their designated certified operator(s).

8. If the facility is a mobile home park, do any residents own their own lot? ☐ Yes ☐ No ☒ NA

9. If applicable, population served by treatment plant (systems that serve the public only) 11300

10. Has the service area of wastewater plant changed since last application submittal? (public system only) ☒ Yes ☐ No ☐ NA

Cadillac Junction development added

11. Identify chemicals used in treatment system

Chemical Name	Use/Purpose	Concentration	Location Feed Point	Authorized
Ferric chloride	Phosphorous removal	38.5%	Aeration tanks	X
Sodium hypochlorite	Periodic treatment of filamentous	90% (tablet)	Various	X

Cadillac WWTP : 15637

NPDES - Facility Site Review

Inspector: Donal Brady

Start Date: 05/17/2018 Completed Date: 09/28/2018

Inspection Information

12. Identify the influent and effluent monitoring points

Parameter	Location	Acceptable
Influent	After headworks screen	X
Effluent	After tertiary filters and after UV	X

13. Number of wastewater personnel recommended in the O&M Manual (public systems)

The manual only discusses shifts not number of personnel.

Specific Processes Producing Wastewater (Industrial/Commercial Only) & Treatment Unit Processes

14. List the specific industrial/commercial processes that produce the raw wastewater flows:

Process Name

This section not completed as WWTP is municipal.

15. Identify the types of wastewater generated.

16. Are these wastewater sources identified in the current permit application?

☐ Yes ☐ No ☒ NA

17. List all treatment unit processes.

activated sludge, clarification

Primary treatment (screening, grit removal, and clarification), activated sludge/secondary clarification, tertiary treatment (RBCs and cloth filters), UV disinfection

18. Are all treatment units in service? If No, explain in comments below, including why, how long out of service, and when expected to be back in service.

☒ Yes ☐ No ☐ NA

General Observations/Questions

19. Operating schedules are established.

☒ Yes ☐ No

20. Number of shifts waste treatment system is in operation.

Shift #	Sun	Mon	Tues	Wed	Thur	Fri	Sat
Shift 1	3 hours	7:00 AM - 3:30 PM	7:00 AM - 3:30 PM	7:00 AM - 3:30 PM	7:00 AM - 3:30 PM	7:00 AM - 3:30 PM	3 hrs
Shift 2							
Shift 3							

21. No. of wastewater treatment personnel

7

7 personnel including Doug

Cadillac WWTP : 15637

NPDES - Facility Site Review

Inspector: Donal Brady

Start Date: 05/17/2018 Completed Date: 09/28/2018

General Observations/Questions

22. How are alarms/upsets monitored and responded to during off hours and on weekends? telephone auto-dialer

Dialer calls City offices between 3:30 and 5:00 PM, after 5:00 PM and on weekend the dialer calls Central Dispatch. WWTP personnel are the subsequently notified by City or Central Dispatch personnel.

23. Does the number of wastewater treatment personnel appear to be adequate? ☒ Yes ☐ No ☐ NA

24. Any bypassing of the collection system or at the WWTP since last inspection? ☒ Yes ☐ No ☐ NA

Bypasses of certain sewer pipes within the collection system occurred during construction projects (bypasses normal for these types of construction projects)

25. If Yes, what is the date of bypass and date notification made to DEQ?

Date of bypass

Date notification made to DEQ

Notifications not required

26. Are hydraulic and/or organic overloads experienced? If Yes, explain in comments. ☐ Yes ☒ No ☐ NA

27. Has there been an increase in flow since your last permit was issued? ☐ Yes ☒ No

Flows have actually decreased due to a reduction in water use

28. Is the current flow rate greater than that authorized by the current permit? ☐ Yes ☒ No

Permit used a design flow of 3.2 MGD

29. is permit modification needed? ☒ Yes ☐ No

The City would like to consider adjusting the seasonal limits in the permit to better account for spring and/or fall weather and the weather's affect on the plant biology

30. Influent flow is measured before all return lines? (public system only) If No, where is it measured? ☒ Yes ☐ No ☐ NA

31. Effluent flow is measured after all return lines? ☐ Yes ☒ No ☐ NA

Some effluent is used for backwashing but the plant accounts for this water

32. Select type of effluent flow metering weir

33. When was it last calibrated? Aug 2017

General Observations/Questions Cont.

34. Has the facility received any citizen complaints or experienced any spills since the last inspection? ☐ Yes ☒ No ☐ NA

Cadillac WWTP : 15637

NPDES - Facility Site Review

Inspector: Donal Brady

Start Date: 05/17/2018 Completed Date: 09/28/2018

General Observations/Questions Cont.

35. Any unusual conditions noted in treatment system effluent, receiving water, irrigation fields, infiltration basins, or tile fields? If Yes, describe in comments section.	☐ Yes	☒ No	☐ NA
36. Any evidence of bypasses or chemical spills, surcharges and/or overflow in the wastewater collection or treatment system? If Yes, describe in comments section.	☐ Yes	☒ No	☐ NA
37. Any accumulation of solids, grease, foam or floating material in wetwells, sumps, clarifiers, land application areas, or other treatment units?	☐ Yes	☒ No	☐ NA
38. Any excessive discoloration, puddles, oil or grease on ground or floor near equipment or tanks?	☐ Yes	☒ No	☐ NA
39. Any problems with wastewater treatment chemical feed system? If Yes, describe in comments section.	☐ Yes	☒ No	☐ NA
40. Do chemical storage areas have adequate curbing and secondary containment to capture leaks or spills?	☒ Yes	☐ No	☐ NA
41. Any jury rigged or temporary systems to correct operation problems? If Yes, describe in comments	☐ Yes	☒ No	☐ NA
42. Any obnoxious odors noted? If Yes, describe from where in comments.	☐ Yes	☒ No	☐ NA
43. Were any unidentified/unmarked confined spaces observed? If Yes, describe where in comments.	☐ Yes	☒ No	☐ NA
44. Is lighting adequate for safe operation and observation?	☒ Yes	☐ No	☐ NA
45. Are aisles, ladders, stairs and other access points kept clear and in good condition?	☒ Yes	☐ No	☐ NA
46. Does general housekeeping appear to be adequate?	☒ Yes	☐ No	☐ NA
47. Does the treatment system security/fencing appear to be adequate?	☒ Yes	☐ No	☐ NA
48. Were photographs taken?	☐ Yes	☒ No	
49. Have all deficiencies previously cited in inspections, enforcement actions, correspondence and verbal communications been corrected?	☐ Yes	☒ No	☐ NA
Effluent BOD is now within permit limits, additional asset management information needs to be submitted			
50. Is a permit modification in order due to a change in name, ownership, or operations?	☐ Yes	☐ No	☒ NA

Closing Information

51. Closing Conference Notes:

DEQ will send draft form and request for additional information

Cadillac WWTP : 15637

NPDES - Facility Site Review

Inspector: Donal Brady

Start Date: 05/17/2018 Completed Date: 09/28/2018

Closing Information

52. Completed By:

Don Brady, PE

Cadillac WWTP : 15637

NPDES - Self-Monitoring Records

Inspector: Donal Brady

Start Date: 05/17/2018 Completed Date: 09/28/2018

General Information

1. Facility Designated Name:	Cadillac Wastewater Treatment Plant
2. Permit Number:	MI0020257
3. Date:	05/17/2018
4. Time:	9:00

Record Information

5. Are all records and information resulting from the monitoring activities required by this permit maintained for at least three years? Examples include DMR's, MORs, lab data, chain of custody forms, calibration & maintenance records and recordings from continuous monitoring instrumentation.	☒ Yes ☐ No ☐ NA
The plant maintains these records for 5 years	

Use the table(s) in the following section to spot check at least three sets of data points (including geometric means, loadings, averages, etc.).

Analytical Tables

6. Use the table below to spot check sets of data points (including geometric means, loadings, averages, etc.).						
Date	Sampling Point:	Type of Wastewater	Parameter	Reported Result (from DMR)	Site Record (from bench sheet or lab)	Accurate?
01/05/2018	Effluent	Municipal	CBOD	6.20	6.20	X
01/08/2018	Effluent	Municipal	Ammonia	2.11	2.11	X
04/20/2018	Effluent	Municipal	CBOD	12.3	12.3	X
7. Are analytical results/bench sheets consistent with the data reported on the DMRs? Explain any discrepancies in the comments.						
☒ Yes ☐ No ☐ NA						

Record Information Cont.

8. Does the facility have authorization for elimination of, or reduced monitoring? If yes, complete the table below.	☐ Yes ☒ No ☐ NA
9. Monitoring Records are adequate and include:	
Parameter	Check if included
Dates, times, frequency and location of sampling collection	X
Name of individual(s) performing sampling	X
Results of analyses	X
Dates of analyses	X
Name of person(s) performing analyses	X
Analytical technique or method used	X
10. Are laboratory bench sheets and other records understandable and clearly filled out?	☒ Yes ☐ No ☐ NA
11. Does the inspection contact/operator have a current copy of the permit?	☒ Yes ☐ No ☐ NA

Cadillac WWTP : 15637

NPDES - Self-Monitoring Records

Inspector: Donal Brady

Start Date: 05/17/2018 Completed Date: 09/28/2018

Record Information Cont.

12. Can the permittee produce their application and WRD approvals for the use of water treatment additives (WTAs)? ☐ Yes ☒ No ☐ NA

WTAs likely approved in accordance with Part 41 permitting for WWTP upgrade construction

Closing Information

13. Closing Conference Notes:

DEQ will send draft form and request for additional information

14. Completed By:

Don Brady, PE

Cadillac WWTP : 15637

NPDES - Sampling Procedures

Inspector: Donal Brady

Start Date: 05/17/2018

Completed Date: 09/28/2018

General Information

1. Facility Designated Name: Cadillac Wastewater Treatment Plant

2. Permit Number: MI0020257

3. Date: 05/17/2018

4. Time: 9:00

5. Sampling Parameters (if answer is not available, use comments section):

Parameter	Sampling Location	Sampling Equipment (auto-sampler, etc)	Sample Type (grab, composite, etc)	Sampling Location Representative?	Sampling Container Material and Quality	Preservation Method	Holding Time
NH3-N	Tertiary filter	Autosampler	Composite	X	HDPE	Cool	28 days
TSS	Tertiary filter	Autosampler	Composite	X	HDPE	Cool	7 days
(C) BOD5	Tertiary filter	Autosampler	Composite	X	HPDE	Cool	48 hrs
Total Phosphorus	Tertiary filter	Autosampler	Composite	X	HPDE	H2SO4, Cool	28 days
TDS	Not in permit						
Fecal Coliform	Outfall		Grab	X	Glass	Cool	8 hours
TRC/TRO	Not in permit						
LL	Not in permit						
Hg	Tertiary filter		Grab	X	Lab supplied	Cool	72 hours
pH	Tertiary filter		Grab	X			15 mins
DO	Tertiary filter		Grab	X			15 mins
O & G	Not in permit						
Metals	Tertiary filter		Grab	X	Lab supplied	NO3, Cool	6 months
Temperature	Not in permit						
WET	Tertiary filter		Grab	X	Lab supplied	Cool	30 hrs
VOCs	Not req'd 2018						
BNAs	Not req'd 2018						
phenols	Not req'd 2018						
Cyanide	Tertiary filter		Grab	X	Lab supplied	Various	14 days

Metals = copper

Inspection Information

6. How are staff trained to collect samples? On-the-job training

7. Are sample collection procedures reviewed periodically? ☒ Yes ☐ No

8. Are sampling procedure changes made when appropriate? ☒ Yes ☐ No

9. Are 24-hour composite samples flow-proportional? If No, explain how it is representative of the discharge in the comments. ☒ Yes ☐ No ☐ NA

Cadillac WWTP : 15637

NPDES - Sampling Procedures

Inspector: Donal Brady

Start Date: 05/17/2018 Completed Date: 09/28/2018

Inspection Information

10. Has an alternate sample type been approved?

Parameter	Sample Type
1	
2	
3	
4	
5	

Not applicable

11. Does the facility have an approved alternate monitoring location?

☐ Yes ☐ No ☒ NA

Composite Sampler Information (1)

12. Monitoring Point Location:

Effluent (after tertiary treatment)

13. Review and/or inspect the composite sampler(s).

Question	Yes	No	NA
Is the temperature in the composite sampler 0-6°C?	X		
Does the sample line maintain a minimum velocity of 2ft/s?	X		
Is a line purging cycle used between samples?	X		
Any obstructions in the sample line?		X	
Does the length or number of bends in the sample line interfere with obtaining a representative sample?		X	

Inspection Information Cont.

14. Is the laboratory sample refrigerator temperature 0-6°C? If not, explain in the comments.

☒ Yes ☐ No ☐ NA

15. Does the facility conduct sampling in accordance with the requirements in the permit's Special Conditions? If No, explain in comments.

☒ Yes ☐ No ☐ NA

16. Is monitoring and analyses performed more often than required by the permit?

☒ Yes ☐ No ☐ NA

More samples are collected/analyzed simply for WWTP process control

17. If Yes, are the results included in the facility's monthly reports?

☐ Yes ☐ No ☒ NA

18. Are 40 CFR Part 136 sampling requirements being met (e.g. Method 1669 for low level mercury)?

☒ Yes ☐ No ☐ NA

Closing Information

19. Closing Conference Notes:

DEQ will send draft form and request for additional information

Cadillac WWTP : 15637

NPDES - Sampling Procedures

Inspector: Donal Brady

Start Date: 05/17/2018 Completed Date: 09/28/2018

Closing Information	
20. Completed By:	Don Brady, PE

Cadillac WWTP : 15637

NPDES - Analytical Methodology

Inspector: Donal Brady

Start Date: 05/17/2018 Completed Date: 09/28/2018

General Information

1. Date:	05/17/2018
2. Time:	9:00 AM
3. Facility Designated Name:	Cadillac Wastewater Treatment Plant
4. Permit Number:	MI0020257

Contract Laboratory

5. Provide the name of any contract lab used for sample analysis and list the parameters analyzed by that lab in the table below.

Name of Lab	Parameter(s)
Pace Analytical	Mercury, copper, cyanide
WET	ERM
HML	Pathogens in biosolids

6. Evaluate the chain of custody record for the following elements:

Facility name, Sample date, Sample location, Sampler name, Number of samples, Parameters to be analyzed, Preservative used, Intake temperature, Signatures, Dates and times

Pace and ERM COCs reviewed

7. Does the contract lab have a QA/QC Program?

☒ Yes ☐ No ☐ NA

8. Does the permittee periodically review the QA/QC program associated with the analyses performed by the contract lab to ensure it is adequate?

☒ Yes ☐ No ☐ NA

The WWTP is familiar with the contract labs due to past experience. The WWTP reviews the contract labs certifications (TNI-NELAP and state certifications).

Analytical Methods Used

9. Analytical Methods Table

Parameter	Method Used	Approved / Acceptable Method	Comments
Fecal coliform	Colilert-18	X	

10. Select one or more parameters to evaluate the protocols and/or techniques used during the analysis to determine if it meets the requirements of the test method.

Parameter	Testing protocols acceptable?
Fecal coliform	X

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Analytical Methods Used

11. Approval has been obtained for alternative test methods? ☐ Yes ☐ No ☒ NA

12. Facility has reference manuals for the test methods that they are using? ☒ Yes ☐ No ☐ NA

WWTP has 22nd and 23rd edition of the Standard Methods for the Examination of Water and Wastewater

13. Facility has written procedures or manuals for instruments and equipment use. ☒ Yes ☐ No ☐ NA

Paper manuals are maintained in the lab, some information is also available in PDF form

14. Are laboratory equipment calibration and maintenance records maintained? ☒ Yes ☐ No ☐ NA

Instrumentation

15. List the analytical instruments the facility uses in their self-monitoring activities. *Confirm acceptability of calibration method using manufacturer specifications or test method requirements

Instrument	Calibration Method*	Frequency	Date last serviced
ISE (ammonia)	Operator cal curve per method	Weekly	Within 1 year
Hach LBOD probe (luminous)(BOD)	Operator cal curve per method	Daily	Within 1 year
Balance	Third party contractor	Annually	Within 1 year
UV Spectrophotometer (P)	Operator cal curve per method	Every 6 weeks	Within 1 year
Flame atomic absorption SP (metals)	Operator cal curve per method	Annually (at a minimum)	Within 1 year

Instruments calibrated within the last year

16. Do laboratory instruments and equipment appear to be in good condition? ☒ Yes ☐ No ☐ NA

17. Are records kept with dates of laboratory equipment maintenance/repair/replacement? ☒ Yes ☐ No ☐ NA

Regular and non-regular maintenance records maintained.

18. Proper volumetric glassware is used? ☒ Yes ☐ No ☐ NA

Appropriate "To contain" and "To deliver" glassware used per methods/procedures

19. Glassware is properly cleaned and sterilized, if necessary? ☒ Yes ☐ No ☐ NA

The lab recently purchased/installed a washer for the glassware

20. Select the type of analytical balance used. If the balance used it "Other", describe it in the comment section. Electronic

Mettler Toledo AG 204

21. Is the balance clean and free of loose powders, stains, and corrosion? ☒ Yes ☐ No ☐ NA

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Instrumentation

22. If the balance is not equipped with automatic vibration compensation, is it mounted on a solid, vibration free surface, isolated from excessive heat, cold, or drafts? ☒ Yes ☐ No ☐ NA

Balance on heavy table

23. Equipment temperature log maintained for ovens, incubators, etc. ☒ Yes ☐ No ☐ NA

The plants BOD incubator and drying oven are equipped with alarms

24. Equipment temperatures. Indicate whether each is acceptable by checking Yes, No, or NA.

Equipment and allowable temperature	Actual temperature	Yes	No	NA
BOD incubator at 20°C? (±1°)	20.2	X		
Fecal coliform incubator at 44.5°C? (±0.2°)	44.8	X		
Drying oven for SS at 103-105°C? (±1°)	105	X		
Muffle furnace at 550°C? (±50°)	550	X		

The WWTP will soon replace the fecal coliform incubator

25. Has the dessicant used with the suspended solids analysis reached its moisture saturation point? ☐ Yes ☒ No ☐ NA

26. Are temperatures verified with a National Institute of Standards and Technology (NIST) thermometer or NIST verified thermometer? ☒ Yes ☐ No ☐ NA

The thermometers are recertified by a third party

Quality Assurance/Quality Control (QA/QC)

27. Are QA/QC written procedures in place for all required monitoring? ☒ Yes ☐ No ☐ NA

28. Are QA/QC results _____ on control charts? N/A

The WWTP will start tracking QA/QC with Excel

29. If data is plotted on control charts, does the data stay within upper and lower control limits? ☐ Yes ☐ No ☒ NA

30. If QA/QC data are outside of control limits, what was done to identify the cause?

Replace reagents,
Check equipment,
Create a new
phosphorus curve,
Clean glassware, Check
laboratory water,
Review testing
protocols, Review data
for accuracy

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Start Date: 05/17/2018 Completed Date: 09/28/2018

Quality Assurance/Quality Control (QA/QC)

31. Were the QA/QC issues adequately addressed?

☒ Yes ☐ No ☐ NA

32. Using the same parameters previously reviewed under the Analytical Methods section, what QA/QC methods are used? If not applicable, put NA in the first parameter box and delete all other pre-populated parameters.

Parameter	Duplicates Frequency	Spikes Frequency	Blank/GGA Frequency	Slope Check	Reference Samples Frequency	Inter-lab Splits Frequency
CBOD (D,GGA,R,I)	1X/week		1X/day		1X/year	1X/year (with Haring)
TSS (D,R,I)	1X/week		1X/week		1X/year	1X/year (with Haring)
Ammonia as N (D,S,Slope,R,I)	1X/week	1X/week	1X/week	1X/day	1X/year	1X/year (with Haring)
Total phosphorus (D,S,R,I)	1X/week	1X/week	1X/week	1X/week & Cal Crv/6 wks	1X/year	1X/year (with Haring)
TRC (D,Slope,R)	Not in permit					
pH (D,Slope,R)	1X/week			Cal Crv 1X/week	1X/year	
Fecal coliform (D,B)	1X/week	1X/week (+ check)	1X/week		1X/year	

33a. If an EPA major or selected minor, has the facility participated in the EPA DMR/QA program?

☒ Yes ☐ No ☐ NA

Facility participated in DMR/QA 37 (2017) and DMR/QA 38 (2018)

33b. If yes, characterize their result from the list below. Use the Comments to explain results, if needed. Acceptable

34. Are there chemicals in the laboratory with an expired shelf life? If the chemical is associated with a parameter being monitored for the NPDES permit, identify the name of the chemical and the expiration date of that chemical in the Comment Section.

☐ Yes ☒ No ☐ NA

Laboratory Data Handling and Personnel

35. Significant figures are established for each analysis and round-off rules are uniformly applied?

☒ Yes ☐ No ☐ NA

Results read off instruments, standards used good to at least 2 significant figures, metals/contract lab results rounded down to 2 significant figures

36. Does staffing appear to be sufficient? If No, provide an explanation in the Comment Section.

☒ Yes ☐ No ☐ NA

37. Is there sufficient time allowed for analyses, QA/QC, laboratory maintenance, and recordkeeping? If No, provide an explanation in the Comment Section.

☒ Yes ☐ No ☐ NA

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Laboratory Data Handling and Personnel

38. How are lab analysts trained?"

On-the-job training

On the job training and job shadowing, followed by a demonstration of capability with method(s), training is renewed/repeated annually

39. Any changes in laboratory personnel since the last inspection? If Yes, identify what the change was and how (if at all) it has affected laboratory operations in the Comment Section.

☐ Yes ☒ No ☐ NA

Cindy is relatively new, but is very experienced, and is working to make incremental improvements/changes to lab

Closing Information

40. Closing Conference Notes:

DEQ to send draft form and request for additional information

41. Completed By:

Don Brady, P.E.

Cadillac WWTP : 15637

NPDES - Municipal/Public Operation and Maintenance Inspection

Inspector: Donal Brady

Start Date: 05/17/2018

Completed Date: 09/28/2018

General Information

Facility Designated Name:

Cadillac Wastewater
Treatment Plant

Permit No.:

MI00020257

Collection System

1. Collection System Categorization

Combined %

Separate %

100

2. Who operates and maintains the collection system?

Contact Name

City of Cadillac - Doug Langworthy and Jeff Dietlin

Telephone Number

(231) 775-2841

3. Is the collection system maintained on a regular schedule?

☒ Yes ☐ No ☐ NA

The collection system is inspected daily

4. Is staffing adequate to maintain system?

☒ Yes ☐ No ☐ NA

5. Has infiltration/inflow been identified as a problem on the collection system?

☐ Yes ☒ No ☐ NA

5A. If yes, explain:

6. Does the collection system or the WWTP have combined sewer overflow retention basins or sanitary sewer overflow equalization basins?

☐ Yes ☒ No ☐ NA

WWTP has 900,000 gallon equalization tank, flow through plant set at 1.75 MGD

6A. If yes, list the basin facilities and state how many times the basins discharged over the past 12 months.

7. Have SSO's occurred in the same general location?

☐ Yes ☒ No ☐ NA

7A. If yes, explain:

8. List the location of pump &/or lift stations used to collect and transport wastewater to the treatment plant.

Station #	Flow (mgd)	Location
Leeson	0.3042	504 Leeson
Leisure	0.224	171 Leisure
Lake	0.3667	601 S Lake
Sunnyside	0.0583	1218 Sunnyside
Cadillac Junction	0.0312	2500 E M-55
Haynes	N/A	336 Haynes
Sixth	0.0078	1000 Sixth St
Roland	0.0824	1090 Roland St
Whaley	0.0956	436 Whaley
13th	N/A	2695 W 13 th St
Weigle	N/A	2703 R.A. Weigle

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NPDES - Municipal/Public Operation and Maintenance Inspection

Inspector: Donal Brady

Start Date: 05/17/2018

Completed Date: 09/28/2018

Collection System

Treatment Plant

1. General types and volumes of wastewater discharged to the municipal system.

Domestic % 60

Non-domestic %

2. Wastewater Treatment Plant Average Influent Strength versus Design.

B.O.D. Influent (mg/L)	T.S.S. Influent (mg/L)	Total Phosphorus Influent (mg/L)	Ammonia Nitrogen Influent (mg/L)	B.O.D. Design (mg/L)	T.S.S. Design (mg/L)	Total Phosphorus Design (mg/L)	Ammonia Nitrogen Design (mg/L)
200	150-200	2.5-3.5	20-30				

3. Does the annual operating budget for the facility appear to be adequate? ☒ Yes ☐ No ☐ NA

4. Has the facility developed an Asset Management Plan? ☒ Yes ☐ No ☐ NA

More detail required for asset management plan and program

5. Number of staff to operate and maintain the wastewater treatment plant? 7

6. Does staffing appear adequate? ☒ Yes ☐ No ☐ NA

7. Are there established procedures for training new operators? ☐ Yes ☒ No ☐ NA

Staff are trained on the job

7A. Describe the training procedures:

On the job training

Does the facility have written procedures for:

8A. Responding to emergency conditions ☒ Yes ☐ No ☐ NA

8B. Reporting unusual/emergency conditions to the DEQ ☒ Yes ☐ No ☐ NA

WWTP follows written procedures in NPDES permit

8C. Laboratory analysis ☒ Yes ☐ No ☐ NA

8D. QA/QC ☒ Yes ☐ No ☐ NA

9. Are the above written procedures adequate? ☒ Yes ☐ No ☐ NA

9A. If no, explain:

Are the following "aids" available to staff and the WWTP superintendent?

10A. Operation and maintenance manual ☒ Yes ☐ No ☐ NA

10B. As built plan and specifications ☒ Yes ☐ No ☐ NA

10C. Equipment manufacturers operation and maintenance manuals ☒ Yes ☐ No ☐ NA

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Inspector: Donal Brady

Start Date: 05/17/2018

Completed Date: 09/28/2018

Treatment Plant

10D. Safety Program

☒ Yes ☐ No ☐ NA

WWTP has a confined space entry/rescue, CPR, and first aid programs

11. Are the alarm systems for equipment and/or power failure or ventilation system adequate?

☒ Yes ☐ No ☐ NA

11A. If no, explain:

12. Does the treatment plant have, or have access to, a backup power supply sufficient to operate the facility and maintain compliance with their NPDES permit?

☒ Yes ☐ No ☐ NA

12A. Please specify the type of backup power supply.

Generator

Backup power for WWTP and large pump stations provided with stationary generators, backup power for small pump stations provided with portable generators

13. Are preventative maintenance activities performed as recommended? Select three to four major equipment items and compare recommended maintenance activities against records kept by the wastewater treatment plant staff.

☒ Yes ☐ No ☐ NA

13A. If no, please explain:

WWTP uses a digital work order system, each month approximately 300 orders are generated, completed orders are entered into the system and uncompleted orders are regenerated the next month, examples of orders include equipment oil changes, equipment greasing, equipment belt changes, order/task frequencies range from every month to every 5 years, a monthly checklist is also completed

14. Does the wastewater treatment plant have an adequate spare parts inventory?

☒ Yes ☐ No ☐ NA

14A. If no, explain:

15. Is the disinfection system adequately operated and maintained?

☒ Yes ☐ No ☐ NA

Pump Stations

1. List the location of pump &/or lift stations used to collect and transport raw wastewater.

Station #

Location

Owned By

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Inspector: Donal Brady

Start Date: 05/17/2018

Completed Date: 09/28/2018

Pump Stations

2. Describe the pump stations (construction type, pumping method, number of pumps, etc.,) by checking one or more of the following:

Parameter	Station 1	Station 2	Station 3	Station 4	Station 5
Capacity					
Prefabricated station (wet well/dry well)					
Built-in-place station					
Submersible pumps					
Pneumatic ejector pumps					
Centrifugal pumps					
Inclined screw (Archimedes) pump					
Elapsed time meter on each pump					
Number of pumps					
On-site alarms (buzzer, flashing light)					
Remote alarms (telemetered)					
Odor complaints received					
Pump Controls (Hand-Off-Auto)					
Water level monitoring (floats/bubblers/sonic)					
Back-up power supply					
Pump around capability					
Other (describe):					

3. Does the maintenance staff keep adequate records of pump station maintenance?

☒ Yes ☐ No ☐ NA

4. How often are the pump stations inspected?

Daily

5. Have there been any problems with grease, rags, or other debris in the wet well, interfering with the proper operation of the level sensor?

☐ Yes ☒ No ☐ NA

Stations are cleaned at least 2X/month, spare pumps are on hand if necessary

5A. If yes, how was it dealt with?

6. When was the last time the pump station wet well was cleaned (to remove grit, grease, scum, rags, etc.)?

May 10th, 2018

7. Any unusual sounds, vibrations or excessive heat noted while the pumps were operating?

☐ Yes ☒ No ☐ NA

7A. If yes, describe:

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Pump Stations

8. Does the alarm system work properly? (Ask staff to conduct a test of the system.)

☒ Yes ☐ No ☐ NA

9. What failures does the alarm system cover?

Power failure, High wet well level

Alarms include high level and power loss, at a minimum

9A. If Other, please describe:

10. Was the following equipment operational? (YES/NO/NA)

Pumps	Water level sensors	Compressors	Check valves	Ventilation	Dehumidifiers
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11. Describe the municipality's procedures/equipment for dealing with a pump station failure (either mechanical or electrical):

Same notification procedures followed for pump station alarm/failure as for WWTP alarms

12. Describe the overall condition of the pump station(s) and note any specific problems:

Closing Notes

Closing Conference Notes:

DEQ to provide draft forms and requests for additional info.

Completed By:

Don Brady, PE

Date of Inspection:

05/17/2018

Cadillac WWTP : 15637

NPDES - Flow Measurement

Inspector: Donal Brady

Start Date: 05/17/2018 Completed Date: 09/28/2018

Inspection Information

1. Facility Designated Name:	Cadillac Wastewater Treatment Plant
2. Permit Number:	MI0020257
3. Date:	05/17/2018
4. Time:	9:00 AM
5. Does the facility have loading limits authorized in their permit?	☒ Yes ☐ No
If No, complete the Type I Outfall section. Definition: These outfalls have no loading limits (e.g. pounds per day, pounds per month) in the NPDES permit.	
If Yes, complete the Type II Outfall or Type III Outfall section. Definition: Type II Outfalls have in-line flow meters and loading limits. Definition: Type III Outfalls have weir and/or flumes and loading limits.	

Type I Outfall

1. What flow measurement device(s) does the facility utilize? Estimation of flows is acceptable through pump curves, water meters, mass balance, use summaries, or similar techniques.		
Outfall Number	Flow Measurement Method	Data Recording Device
2. Does the facility use pump curves or pump ratings? ☐ Yes ☐ No ☐ NA		
2a. How old is the pump?		
2b. How are pumping times obtained?		
2c. When was the pump curve last calibrated/checked?		
2d. Who calculated the pump curve?		
2e. Have there been any maintenance problems with the pump? If yes, describe in the comments. ☐ Yes ☐ No ☐ NA		
2f. Does the pumping rate vary significantly during discharge periods? ☐ Yes ☐ No ☐ NA		
2g. If yes, is a head-discharge curve used? ☐ Yes ☐ No ☐ NA		
2h. Do the conversion calculations for changing pump times to flow make sense? ☐ Yes ☐ No ☐ NA		
2i. Are pressure gauges installed, read and recorded? ☐ Yes ☐ No ☐ NA		
2j. If pressure gauges are used, is the height from the water surface to the gauge added to the gauge reading for total head? ☐ Yes ☐ No ☐ NA		
3. Are water meters used? ☐ Yes ☐ No ☐ NA		

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NPDES - Flow Measurement

Inspector: Donal Brady

Start Date: 05/17/2018

Completed Date: 09/28/2018

Type I Outfall

3a. When was the meter installed?

3b. Is the meter ever calibrated against an alternate method of flow measurement at least annually?

☐ Yes ☐ No ☐ NA

3c. Is the meter installed such that it remains full of liquid?

☐ Yes ☐ No ☐ NA

4. Are usage summaries used?

☐ Yes ☐ No ☐ NA

4a. When was the summary compiled?

4b. Who developed the summary?

4c. Have changes occurred in plant operations since the summary was compiled?

☐ Yes ☐ No ☐ NA

4d. Has the summary ever been compared against an alternate method of flow measurement?

☐ Yes ☐ No ☐ NA

5. Is the appropriate flow measurement method used for all Type I Outfalls? If not, describe which outfall and why it is not acceptable in the comments.

☐ Yes ☐ No ☐ NA

Type II Outfall

1. What flow measurement device(s) does the facility utilize? It is acceptable to use magnetic meters, doppler meters, venturi meters, flow nozzle meters, and orifice meters.

Outfall Number

Flow Measurement Method

Data Recording Device

2. Does the facility use a magnetic meter(s)? Magnetic Meters - A voltage is created when the flow passes through an electromagnetic field. The voltage is proportional to the velocity of the flow (Faraday's Law.)

☐ Yes ☐ No ☐ NA

2a. Is the flow velocity in the pipe at least three to five feet per second?

☐ Yes ☐ No ☐ NA

2b. Are there any changes in the conductivity of the wastewater during the day (variations in dissolved solids - particularly chlorides?)

☐ Yes ☐ No ☐ NA

3. Does the facility use a doppler meter(s)? Doppler Meters - When an ultrasonic wave is reflected from bubbles, suspended solids, and turbulence in a moving fluid, the frequency shift of the reflected wave will be in proportion to the velocity of the fluid.

☐ Yes ☐ No ☐ NA

3a. If yes, are there significant changes in the suspended solids contents in the wastewater?

☐ Yes ☐ No ☐ NA

4. Does the facility use a venturi, flow nozzle or orifice meter(s)? Venturi, Flow Nozzle and Orifice Meters - The flow rate is determined by the pressure differential across the meter.

☐ Yes ☐ No ☐ NA

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NPDES - Flow Measurement

Inspector: Donal Brady

Start Date: 05/17/2018

Completed Date: 09/28/2018

Type III Outfall

1. What flow measurement device(s) does the facility utilize? It is acceptable to have either weirs or flumes. Liquid flows over weirs, but through flumes.

Outfall Number	Flow Measurement Method	Data Recording Device
001A	Notched weir	Levels transmitter/SCADA

2. Does the facility use a weir(s) for flow measurement? ☒ Yes ☐ No ☐ NA

2a. What type of weir is used? v-notch

2b. Is there excessive sediment build-up behind the weir plate? ☐ Yes ☒ No ☐ NA

2c. Is there leaking around or under the weir plate? ☐ Yes ☒ No ☐ NA

2d. Has debris (algae, sticks, paper) collected on the weir crest? ☐ Yes ☒ No ☐ NA

2e. Is the weir plate perpendicular to the direction of the flow? ☒ Yes ☐ No ☐ NA

2f. Does the weir plate appear level? ☒ Yes ☐ No ☐ NA

2g. Does the weir plate appear to be plumb? ☒ Yes ☐ No ☐ NA

2h. Does the water flow freely and clearly over the weir crest (can an air space be seen below the nappe of the water?) ☒ Yes ☐ No ☐ NA

2i. Is the upstream flow channel straight for at least four times the depth of the water level in the channel? ☒ Yes ☐ No ☐ NA

2j. Is the distance from the weir opening to the side of the channel at least two times the head flowing over the weir? ☒ Yes ☐ No ☐ NA

2k. Does the velocity of approach in the channel cause turbulence near the weir? ☐ Yes ☒ No ☐ NA

2l. How are head measurements taken? Ultrasonic level transmitter

2m. If a bubbler tube or an ultrasonic level measuring device is used, how is the device calibrated? Third party contractor

2n. How often is it calibrated? Annually

2o. What flow measurement formula is the permittee using? $Q \text{ (cfs)} = 3.367 \times [LH \text{ (in)}]^{3/2}$

3. Does the facility use a flume(s) for flow measurement? ☐ Yes ☒ No ☐ NA

3a. Is the channel approach to the flume straight? ☐ Yes ☐ No ☒ NA

3b. Is the flow depth across the channel uniform? ☐ Yes ☐ No ☒ NA

3c. Are there any waves or turbulent conditions in the approach channel? ☐ Yes ☐ No ☒ NA

Cadillac WWTP : 15637

NPDES - Flow Measurement

Inspector: Donal Brady

Start Date: 05/17/2018

Completed Date: 09/28/2018

Type III Outfall

3d. Where is the level (head) measurement taken?

3e. How is the level measured?

3f. If a float device is used, is a stilling well used?

☐ Yes ☐ No ☒ NA

3g. If a bubbler tube, or ultrasonic device is used, how is the device calibrated?

3h. How often is it calibrated?

3i. Does the water flow freely through the throat of the flume?

☐ Yes ☐ No ☒ NA

3j. If the flow does not appear to be free, how does the permittee measure head?

3k. What flow measurement formula is the permittee using?

Closing Information

Closing Comments:

DEQ will send draft form and request for additional information

Completed By:

Don Brady, PE

Date of Inspection:

05/17/2018

Cadillac WWTP : 15637

NPDES - Unit Processes Checklist

Inspector: Donal Brady

Start Date: 05/17/2018 Completed Date: 09/28/2018

Unit Processes Checklist	
<i>1. Facility Name:</i>	Cadillac Wastewater Treatment Plant
<i>2. Preliminary Treatment</i>	Flow Equalization Basin, Bar Screen, Mechanical Grit Chamber
<i>3. Primary Clarifier</i>	Circular
<i>4. Treatment Types</i>	Activated Sludge, Rotating Biological Contractor
<i>5. Secondary Clarifiers</i>	Circular
<i>6. Disinfection</i>	Ultra-Violet Radiation
<i>7. Sludge Storage</i>	Sludge Storage Tank
<i>8. Sludge Thickening</i>	Gravity Thickeners
<i>9. Sludge Dewatering</i>	
<i>10. Sludge Stabilization</i>	Anaerobic Digestion
<i>11. Tertiary Treatment</i>	
<i>12. Tertiary Treatment (Specify Treatment Type)</i>	RBCs and Cloth filters

Cadillac WWTP : 15637

NPDES - Areas Evaluated

Inspector: Donal Brady

Start Date: 05/17/2018

Completed Date: 09/28/2018

Areas Evaluated	
1. Biosolids/Sludge Disposal	Not Evaluated
2. Compliance Schedule	Marginal
Additional asset management information is required	
3. Effluent/Receiving Waters	Satisfactory
4. Facility Site Review	Satisfactory
5. Flow Measurement	Satisfactory
6. Laboratory	Satisfactory
7. Operation & Maintenance	Satisfactory
8. Permit Review	Satisfactory
9. Records/Reporting	Satisfactory
10. Sample Protocol	Satisfactory
11. Self-Monitoring Program	Satisfactory
12. Other	Not Evaluated

ATTACHMENT NAME:

2018 Cadillac WWTP CSI Final Report.xlsx

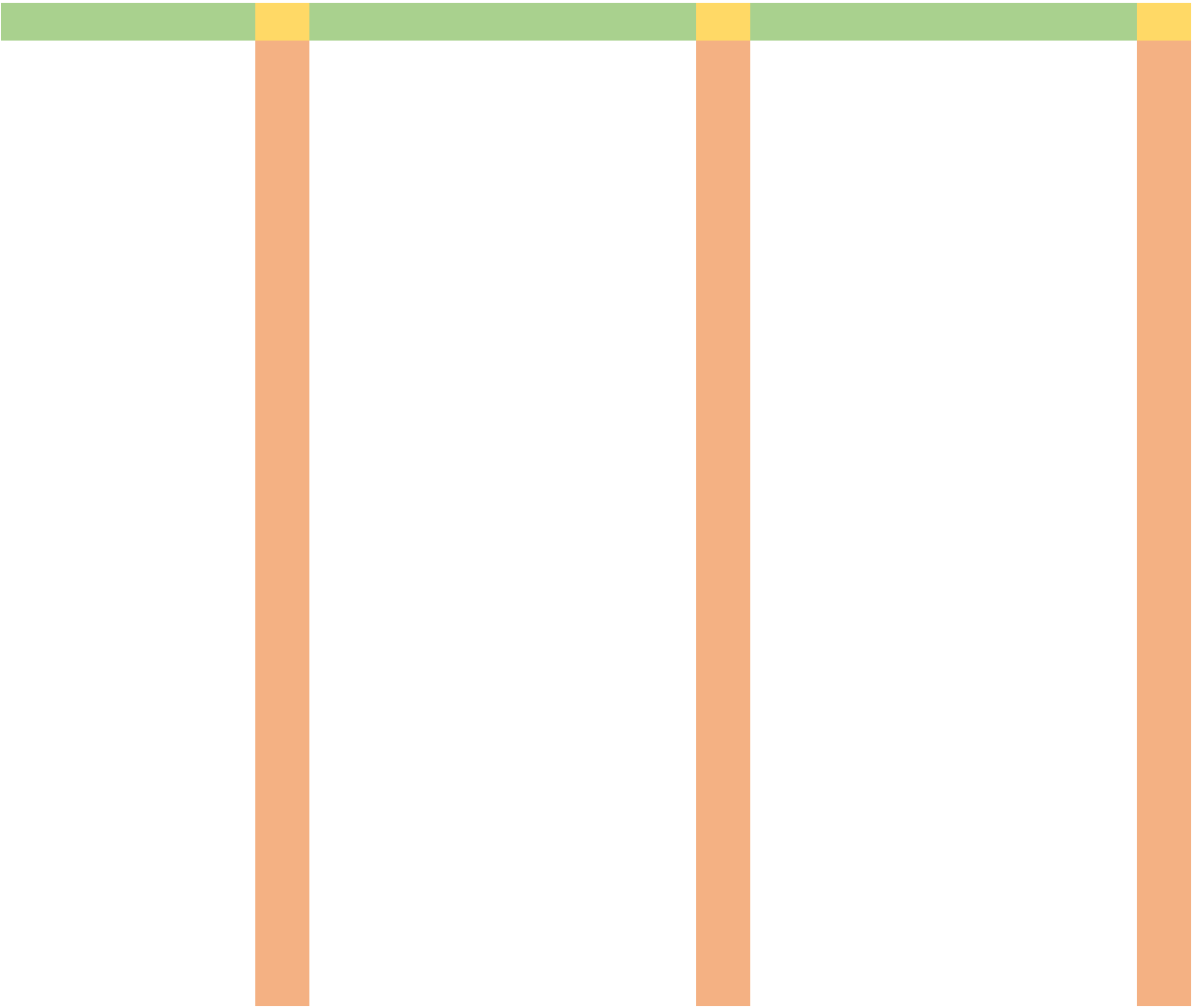
ATTACHMENT TYPE:

Microsoft Office 12+, Office Open XML (OOXML/OpenXML): spreadsheet (XLSX)

Water Composite 1		Det Lev	Water Grab 1 -1
Facility Name	Cadillac WWTP		Cadillac WWTP
Monitoring Point	001A		001A
Survey Period Start Date/time	5/9/2018 10:20		
Survey Period End Date/time	5/10/2018 9:20		
Flow Rate (MGD)	1.975		
Grab Matrix			Wastewater
Grab Date Collected			5/9/2018
Grab Time Collected			10:20
Parameters	Results		Results
Temperature (F)			61
pH (S.U.)			6.99
TRC			
Dissolved Oxygen			8.06
Conductivity (umho/cm)			
Fecal Coliform			---
Oil and Grease			
	Results	Units	Results
Suspended Solids	7	mg/L	<4 ND
Dissolved Solids		mg/L	<20 ND
CBOD-5		mg/L	<3 ND
BOD-5		mg/L	
COD		mg/L	
TOC		mg/L	
NO2		mg/L	<0.01 ND
NO3 + NO2		mg/L	<0.01 ND
NH3	0.12	mg/L	<0.01 ND
Kjeldahl Nitrogen		mg/L	<0.1 ND
Total Phosphorus	0.15	mg/L	<0.01 ND
Ortho Phosphate		mg/L	<0.01 ND
Total Cyanide		µg/L	<5 ND
Amenable Cyanide		µg/L	
Available Cyanide		mg/L	<0.002 ND
Alkalinity (as CaCO3)	130	mg/L	<20 ND
Hardness (as CaCO3)		mg/L	<1 ND
Calcium		mg/L	<1 ND
Sodium		mg/L	<1 ND
Potassium		mg/L	<0.1 ND
Magnesium		mg/L	<1 ND
Chloride		mg/L	<1 ND
Sulfate		mg/L	<2 ND
Sulfide		mg/L	

	Location	Concentration	Unit	Standard
x	Total Arsenic (As)	ND	µg/L	<1 ND
x	Total Barium (Ba)	23	µg/L	<5 ND
x	Total Cadmium (Cd)	ND	µg/L	<0.2 ND
x	Total Chromium (Cr)	1.2	µg/L	<1 ND
x	Total Copper (Cu)	6.4	µg/L	<1 ND
x	Total Lead (Pb)	ND	µg/L	<1 ND
x	Total Mercury (Hg)	ND	µg/L	<0.2 ND
x	Total Selenium (Se)	ND	µg/L	<1 ND
x	Total Silver (Ag)	ND	µg/L	<0.2 ND
x	Total Zinc (Zn)	20	µg/L	<5 ND
	Total Aluminum (Al)		µg/L	<5 ND
	Total Antimony (Sb)		µg/L	<1 ND
	Total Beryllium (Be)		µg/L	<1 ND
	Total Boron (B)		µg/L	<20 ND
	Total Cobalt (Co)		µg/L	<5 ND
	Hexavalent Chromium (Cr+6)		µg/L	<5 ND
	Total Iron (Fe)		µg/L	<20 ND
	Total Lithium (Li)		µg/L	<10 ND
x	Total Nickel (Ni)	4.5	µg/L	<2 ND
	Total Manganese (Mn)		µg/L	<5 ND
x	Total Molybdenum (Mo)	12	µg/L	<5 ND
	Total Strontium (Sr)		µg/L	<5 ND
	Total Thallium (Tl)		µg/L	<2 ND
	Total Titanium (Ti)		µg/L	<5 ND
	Total Vanadium (V)		µg/L	<2 ND
	Phenolics		µg/L	

Det Lev	Water Grab 1 - 2	Det Lev	Water Grab 1 - 3	Det Lev
	Cadillac WWTP 001A		Cadillac WWTP 001A	
	Wastewater		Wastewater	
	5/9/2018		5/10/2018	
	15:35		9:20	
Units	Results	Units	Results	Units
	59		62	
	6.8		8.17	
	8.44		8.33	
(cts/100 ml) <10 ND	---	<10 ND	ND	<10 ND
Units	Results	Units	Results	Units



Upstream	Det Lev
Cadillac WWTP Upstream	

Surface Water
5/9/2018
11:00

Results	Units
62	
7.06	
9.05	

Results	Units

56
71



***MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY
WATER RESOURCES DIVISION
POINT SOURCE MONITORING***

Report of a
Municipal Wastewater Survey
Conducted at
Cadillac WWTP
NPDES No. MI0020257
Wexford County
Cadillac, MI 49601
May 9-10, 2018

SURVEY SUMMARY

Wastewater monitoring was performed during one, twenty-four hour survey period starting May 9, 2018 at 10:20 AM.

The results of the physical, chemical and bacteriological analyses are presented in Tables 1 & 2.

The results of this survey are compared to the results reported by the facility in Tables 1 & 2.

A composite sample from monitoring point 001A was split with the facility to compare laboratory analyses. The results are shown in Table 3.

The results of this survey are compared to the results of the previous survey performed in October of 2014 in Table 4.

Samples were preserved according to Table 5. Letter codes for laboratory results are defined in Table 5.

SURVEY PROCEDURE

The flow and samples were obtained as follows:

<u>Sample Location</u>	<u>Flow Measurement</u>	<u>Sampling Methods</u>
001A - Composite samples collected prior to disinfection from facility compositor	Facility Analysis Report	Grab composite samples
001A - Grab samples collected post disinfection from outfall to Clam River	---	Individual grab samples
Upstream samples collected from approximately 20 ft. upstream of outfall to Clam River	---	Individual grab samples

A grab composite consists of a series of individual grabs composited into one sample.

An individual grab is a single instantaneous sample.

Samples were analyzed by the MDEQ Laboratory located in Lansing, Michigan, and Merit Laboratories, Inc. in East Lansing, MI.

	MDEQ Analyses		Cadillac WWTP Analyses	
Monitoring Point	001A		001A	
Survey Period Start Date/Time	5/9/18 10:20		5/9/18 7:00	
Survey Period End Date/Time	5/10/18 9:20		5/10/18 7:00	
Flow Rate (MGD)	2.14		2.14	
	Results		Results	
	<u>mg/L</u>	<u>lbs/day</u>	<u>mg/L</u>	<u>lbs/day</u>
Suspended Solids	7	124.933	4	71.390
CBOD-5	<3 ND	NA	9.31	166.161
NH3	0.12	2.142	0.556	9.923
Total Phosphorus	0.15	2.677	0.111	1.981
Available Cyanide	<0.002 ND	NA	---	---
Alkalinity (as CaCO3)	130	2320.188	---	---
	<u>µg/L</u>	<u>lbs/day</u>	<u>µg/L</u>	<u>lbs/day</u>
Total Arsenic (As)	<1 ND	NA	---	---
Total Barium (Ba)	23	0.410	---	---
Total Cadmium (Cd)	<0.2 ND	NA	---	---
Total Chromium (Cr)	1.2	0.021	---	---
Total Copper (Cu)	6.4	0.114	---	---
Total Lead (Pb)	<1 ND	NA	---	---
Total Mercury (Hg)	<0.2 ND	NA	---	---
Total Selenium (Se)	<1 ND	NA	---	---
Total Silver (Ag)	<0.2 ND	NA	---	---
Total Zinc (Zn)	20	0.357	---	---
Total Nickel (Ni)	4.5	0.080	---	---
Total Molybdenum (Mo)	12	0.214	---	---

Monitoring Point Matrix Date Collected Time Collected	MDEQ Analyses				Cadillac WWTP Analyses	
	Grab 1-1	Grab 1-2	Grab 1-3	Upstream	Day 1	Day 2
	001A	001A	001A	Upstream	001A	001A
	Wastewater	Wastewater	Wastewater	Surface Water	Wastewater	Wastewater
	5/9/2018	5/9/2018	5/10/2018	5/9/2018	5/9/2018	5/10/2018
	10:20	15:35	9:20	11:00	7:00	7:00
			<u>Results</u>		<u>Results</u>	
Temperature (F)	61	59	62	62	---	---
pH (S.U.)	6.99	6.8	8.17	7.06	7.08	6.94
Dissolved Oxygen (mg/L)	8.06	8.44	8.33	9.05	8.02	7.65
Fecal Coliform (cts/100 ml)	---	---	<10 ND	---	1	<1 ND
Available Cyanide (mg/L)	---	---	<0.002 ND	---	---	---

Composite Split

Monitoring Point

001A

Date Collected

5/10/2018

Time Collected

9:20

	MDEQ Results	Facility Results
	<u>mg/L</u>	<u>mg/L</u>
Suspended Solids	6	4
NH3	0.44	0.556
Total Phosphorus	0.16	0.111
Alkalinity (as CaCO3)	130	---
	<u>µg/L</u>	<u>µg/L</u>
Total Arsenic (As)	<1 ND	---
Total Barium (Ba)	24	---
Total Cadmium (Cd)	<0.2 ND	---
Total Chromium (Cr)	1.1	---
Total Copper (Cu)	7.6	---
Total Lead (Pb)	<1 ND	---
Total Mercury (Hg)	<0.2 ND	---
Total Selenium (Se)	<1 ND	---
Total Silver (Ag)	<0.2 ND	---
Total Zinc (Zn)	20	---
Total Nickel (Ni)	4.7	---
Total Molybdenum (Mo)	13	---

Monitoring Point	001A	
Matrix	Wastewater	
Survey Period Start Date/Time	10/14/14 10:15	5/9/18 10:20

Results

	<u>mg/L</u>	<u>mg/L</u>
Suspended Solids	7	7
CBOD-5	4.8	<3 ND
NH3	0.31	0.12
Total Phosphorus	0.15	0.15
Available Cyanide	---	<0.002 ND
Alkalinity (as CaCO3)	130	130
	<u>µg/L</u>	<u>µg/L</u>
Total Arsenic (As)	---	<1 ND
Total Barium (Ba)	---	23
Total Cadmium (Cd)	---	<0.2 ND
Total Chromium (Cr)	---	1.2
Total Copper (Cu)	7.3	6.4
Total Lead (Pb)	---	<1 ND
Total Mercury (Hg)	---	<0.2 ND
Total Selenium (Se)	---	<1 ND
Total Silver (Ag)	---	<0.2 ND
Total Zinc (Zn)	---	20
Total Nickel (Ni)	---	4.5
Total Molybdenum (Mo)	---	12

Parameter	Preservative
COD/TOC/Phenol/Nutrients (Chlorine Absent)	5 drops H ₂ SO ₄ /250 ml (to pH <2)
Available Cyanide	Preserved with Sodium Hydroxide to a minimum pH of 12.
pH & Dissolved Oxygen (D.O.)	Meter reading
Total Metals	2 ml 1:1 HNO ₃ /250 ml (to pH <2).
Microbiology	2 drops 10% sodium thiosulfate/125 ml to dechlorinate sample

Samples preserved as required, cooled to 6 degrees Centigrade with chain of custody maintained

Lab Letter Codes

ND - Indicates compound analyzed for but not detected

Survey by: Micky Leonard, Environmental Quality Analyst

Contact with Management: Doug Langworthy, Superintendant Cadillac WWTP

Laboratory Analyses by: Michigan Department of Environmental Quality Laboratory,
Lansing and Michigan Merit Laboratories Inc., East Lansing,
Michigan

Report by: Micky Leonard, Environmental Quality Analyst
Field Operations Section
Water Resources Division
Southeast Michigan District Office

Publish Date: 6/15/2018

ATTACHMENT NAME:

Basis for Decision Memo-DRAFT_Cadillac WWTP.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

BASIS FOR DECISION MEMO

Permit Processor: Bradley Popovich
 Date: August 2, 2018
 Permit No. MI0020257
 Facility's Designated Name: Cadillac WWTP

Monitoring Point 001A – Current Seasons: Authorization to discharge 3.2 MGD of treated municipal wastewater from Monitoring Point 001A through Outfall 001. Outfall 001 discharges to the Clam River.

	<u>Maximum Limits for Quantity or Loading</u>				<u>Maximum Limits for Quality or Concentration</u>				<u>Monitoring Frequency</u>	<u>Sample Type</u>	<u>Basis for Limits</u>
<u>Parameter</u>	<u>Monthly</u>	<u>7-Day</u>	<u>Daily</u>	<u>Units</u>	<u>Monthly</u>	<u>7-Day</u>	<u>Daily</u>	<u>Units</u>			
Flow	(report)	---	(report)	MGD	---	---	---	---	Daily	Report Total Daily Flow	PWJ
Carbonaceous Biochemical Oxygen Demand (CBOD ₅)											
May – November	190	270	(report)	lbs/day	7	---	10	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
December – March	670	910	(report)	lbs/day	25	---	34	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
April	450	670	(report)	lbs/day	17	---	25	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
Total Suspended Solids (TSS)											
May – November	530	800	(report)	lbs/day	20	30	(report)	mg/l	5x Weekly	24-Hr Composite	PWJ
December – April	800	1200	(report)	lbs/day	30	45	(report)	mg/l	5x Weekly	24-Hr Composite	PWJ
Ammonia Nitrogen (as N)											
May – November	24	53	(report)	lbs/day	0.9	---	2.0	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
December – March	99	---	(report)	lbs/day	3.7	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
April	130	---	(report)	lbs/day	4.8	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
Total Phosphorus (as P)											
May – March	13	---	(report)	lbs/day	0.5	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL
April	17	---	(report)	lbs/day	0.65	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL
Fecal Coliform Bacteria	---	---	---	---	200	400	(report)	cts/100 ml	5x Weekly	Grab	WQS
Acute Toxicity	---	---	---	---	---	---	(report)	TU _A	Quarterly	24-Hr Composite	
Chronic Toxicity	---	---	---	---	1.1	---	---	TU _C	Quarterly	24-Hr Composite	
Available Cyanide	0.16	---	---	lbs/day	5.9	---	(report)	ug/l	Monthly	Grab	
Total Copper	0.3	---	0.82	lbs/day	11	---	31/(report)	ug/l	Weekly Quarterly	24-Hr Composite	WQBEL

<u>Parameter</u>	<u>Monthly</u>		<u>Daily</u>	<u>Units</u>	<u>Monthly</u>		<u>Daily</u>	<u>Units</u>	<u>Monitoring Frequency</u>	<u>Sample Type</u>	<u>Basis for Limits</u>
Total Nickel	1.4	---	(report)	lbs/day	51	---	(report)	ug/l	Weekly	24-Hr Composite	WQBEL
Total Selenium	0.15	---	(report)	lbs/day	5.7	---	(report)	ug/l	Weekly	24-Hr Composite	WQBEL
Bis(2-ethylhexyl)phthalate	0.82	---	(report)	lbs/day	31	---	(report)	ug/l	Weekly	24-Hr Composite	WQBEL
Total Mercury	(report)	---	---	lbs/day	(report)	---	---	ng/l	Quarterly	Grab	
Corrected	(report)	---	(report)	lbs/day	(report)	---	(report)	ng/l	Quarterly	Calculation	WQC
Uncorrected	---	---	---	---	---	---	(report)	ng/l	Quarterly	Grab	WQC
Field Duplicate	---	---	---	---	---	---	(report)	ng/l	Quarterly	Grab	WQC
Field Blank	---	---	---	---	---	---	(report)	ng/l	Quarterly	Preparation	WQC
Laboratory Method Blank	---	---	---	---	---	---	(report)	ng/l	Quarterly	Preparation	WQC
	12-Month Rolling Avg				12-Month Rolling Avg						
Total Mercury	0.000053	---	---	lbs/day	2.0	---	---	ng/l	Quarterly	Calculation	WQV
					Minimum % Monthly		Minimum % Daily				
TSS Minimum % Removal											
December – April	---	---	---	---	85	---	(report)	%	Monthly	Calculation	STS
					Minimum Daily		Maximum Daily				
pH	---	---	---	---	6.5	---	9.0	S.U.	Daily	Grab	WQS
Dissolved Oxygen											
May – November	---	---	---	---	6.0	---	---	mg/l	5x Weekly	Grab	WQBEL/AD
December – April	---	---	---	---	5.0	---	---	mg/l	5x Weekly	Grab	WQBEL/AD

Monitoring Point 001A – Alternate Seasons: Authorization to discharge 3.2 MGD of treated municipal wastewater from Monitoring Point 001A through Outfall 001. Outfall 001 discharges to the Clam River.

	<u>Maximum Limits for Quantity or Loading</u>				<u>Maximum Limits for Quality or Concentration</u>				<u>Monitoring Frequency</u>	<u>Sample Type</u>	<u>Basis for Limits</u>
<u>Parameter</u>	<u>Monthly</u>	<u>7-Day</u>	<u>Daily</u>	<u>Units</u>	<u>Monthly</u>	<u>7-Day</u>	<u>Daily</u>	<u>Units</u>			
Flow	(report)	---	(report)	MGD	---	---	---	---	Daily	Report Total Daily Flow	PWJ
Carbonaceous Biochemical Oxygen Demand (CBOD ₅)											
June – September <i>May – November</i>	190	270	(report)	lbs/day	7	---	10	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
October – November	290	480	(report)	lbs/day	11	---	17	mg/l	5x Weekly	24-Hr Composite	PWJ
December – March	670	910	(report)	lbs/day	25	---	34	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
April – May / April	450	670	(report)	lbs/day	17	---	25	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
Total Suspended Solids (TSS)											
June – September <i>May – November</i>	530	800	(report)	lbs/day	20	30	(report)	mg/l	5x Weekly	24-Hr Composite	PWJ
October – May <i>December – April</i>	800	1200	(report)	lbs/day	30	45	(report)	mg/l	5x Weekly	24-Hr Composite	PWJ
Ammonia Nitrogen (as N)											
June – September <i>May – November</i>	24	53	(report)	lbs/day	0.9	---	2.0	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
October – November	88	---	(report)	lbs/day	1.6	---	2.8	mg/l	5x Weekly	24-Hr Composite	PWJ
December – March	99	---	(report)	lbs/day	3.7	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
April – May / April	130	---	(report)	lbs/day	4.8	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
Total Phosphorus (as P)											
May – March	13	---	(report)	lbs/day	0.5	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL
April	17	---	(report)	lbs/day	0.65	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL
Fecal Coliform Bacteria	---	---	---	---	200	400	(report)	cts/100 ml	5x Weekly	Grab	WQS
<i>Acute Toxicity</i>	---	---	---	---	---	---	(report)	TU _A	Quarterly	24-Hr Composite	
<i>Chronic Toxicity</i>	---	---	---	---	1.1	---	---	TU _C	Quarterly	24-Hr Composite	

<u>Parameter</u>	<u>Monthly</u>			<u>Units</u>	<u>Monthly</u>		<u>Daily</u>	<u>Units</u>	<u>Monitoring Frequency</u>	<u>Sample Type</u>	<u>Basis for Limits</u>
Available Cyanide	0.16	---	---	lbs/day	5.9	---	(report)	ug/l	Monthly	Grab	
Total Copper	0.3	---	0.82	lbs/day	11	---	31/(report)	ug/l	Weekly Quarterly	24-Hr Composite	WQBEL
Total Nickel	1.4	---	(report)	lbs/day	51	---	(report)	ug/l	Weekly	24-Hr Composite	WQBEL
Total Selenium	0.15	---	(report)	lbs/day	5.7	---	(report)	ug/l	Weekly	24-Hr Composite	WQBEL
Bis(2-ethylhexyl)phthalate	0.82	---	(report)	lbs/day	31	---	(report)	ug/l	Weekly	24-Hr Composite	WQBEL
Total Mercury	(report)	---	---	lbs/day	(report)	---	---	ng/l	Quarterly	Grab	
Corrected	(report)	---	(report)	lbs/day	(report)	---	(report)	ng/l	Quarterly	Calculation	WQC
Uncorrected	---	---	---	---	---	---	(report)	ng/l	Quarterly	Grab	WQC
Field Duplicate	---	---	---	---	---	---	(report)	ng/l	Quarterly	Grab	WQC
Field Blank	---	---	---	---	---	---	(report)	ng/l	Quarterly	Preparation	WQC
Laboratory Method Blank	---	---	---	---	---	---	(report)	ng/l	Quarterly	Preparation	WQC
	12-Month Rolling Avg				12-Month Rolling Avg						
Total Mercury	0.000053	---	---	lbs/day	2.0	---	---	ng/l	Quarterly	Calculation	WQV
					Minimum % Monthly		Minimum % Daily				
TSS Minimum % Removal											
October – May December – April	---	---	---	---	85	---	(report)	%	Monthly	Calculation	STS
					Minimum Daily		Maximum Daily				
pH	---	---	---	---	6.5	---	9.0	S.U.	Daily	Grab	WQS
Dissolved Oxygen											
May – November	---	---	---	---	6.0	---	---	mg/l	5x Weekly	Grab	WQBEL/AD
December – April	---	---	---	---	5.0	---	---	mg/l	5x Weekly	Grab	WQBEL/AD

PERMIT CONDITIONS:

Final Effluent Limitations, Monitoring Point 001A

Quantification Levels and Analytical Methods for Selected Parameters

Additional Monitoring Requirements

Pollutant Minimization Program for Total Mercury

Untreated or Partially Treated Sewage Discharge Reporting and Testing Requirements

Facility Contact

Monthly Operating Reports

Asset Management

Discharge Monitoring Report – Quality Assurance Study Program

Schedule of Compliance (not required)

Storm Water Pollution Prevention (not required)

Michigan Industrial Pretreatment Program

Residuals Management Program for Land Application of Biosolids

NOTES:

The WQBEL Toxic memo recommended monitoring for the following parameters: 3,3-Dichlorobenzidine, Benzidine, Hexachlorobenzene, Hexachlorobutadiene, Hexachlorocyclopentadiene, Acrylonitrile, 4-Chloro-3-methylphenol, Fluoranthene, Pentachlorophenol, Phenanthrene, and 2,4,6-Trichlorophenol. These parameters have not been included in the draft permit because they are not expected in the discharge, however, the compounds and appropriate quantification levels (QL's) are now listed in Part I.A.2, and these parameters are included in the additional monitoring requirements under Part I.A.3.

Less stringent limits are recommended for CBOD₅, Ammonia Nitrogen (as N), and Dissolved Oxygen for Monitoring Point 001A – Current Seasons. Based on a review of the facility's existing effluent quality, Permits Section determined that it is appropriate to retain the current limits based on antidegradation.

Less stringent limits are recommended for Dissolved Oxygen for Monitoring Point 001A – Alternate Seasons. Based on a review of the facility's existing effluent quality, Permits Section determined that it is appropriate to retain the current limits based on antidegradation.

Less stringent limits are recommended for CBOD₅, and Ammonia Nitrogen (as N) for the months of December through September for Monitoring Point 001A – Alternate Seasons. Based on a review of the facility's existing effluent quality, Permits Section determined that it is appropriate to retain the current limits based on antidegradation.

The facility is in the process of applying for a No Exposure Certification (NEC) that will cover any storm water discharges during this permit reissuance cycle.

The requirement to report total mercury results and associated data for corrected, uncorrected, field duplicate, field blank, and laboratory method blank results is added in order to be consistent with the sampling and QA/QC requirements of EPA Method 1631, Revision E, "*Mercury in Water by Oxidation, Purge and Trap, and Cold Vapor Atomic Fluorescence Spectrometry*".

Permits Section followed up with the facility and the facility confirmed that they are no longer using chlorine for disinfection.

An analysis of the facility's existing data was conducted, and it was determined that 99% of the time the facility is meeting the daily PWJ limits for CBOD₅ and Ammonia Nitrogen (as N) for the months of October through November. The completed analysis also found that 95% of the time the facility is meeting the monthly PWJ limit for Ammonia Nitrogen (as N) for the months of October through November).

Designated Name: Cadillac WWTP
Basis for Decision Memo
Page 6

Limit Change Key

Normal Type = existing requirement - carried over from previous version of permit

Bold Type = new requirement - not in previous version of permit

Italic = deleted requirement - not carried over from previous version of permit

Basis for Limits Key

WQBEL - Water Quality-Based Effluent Limit

STS - Secondary Treatment Standard

WQC - Water Quality Concern

WQS - Water Quality Standard

WQV - Water Quality Variance

PWJ - Permit Writer's Judgment

AD – Antidegradation

ATTACHMENT NAME:

Cadillac Asset Mngmnt Submittals 2016 2017 2018.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

Our AMP schedule is:

Current Work

Completing final data collection and Condition Assessment

Performing Level of Service and

Risk and Consequence of Failure Analysis

Summer 2016

Writing Asset Management Plan

Fall 2016 (9/30/16 – 11/30/16)

Completing Asset Management Plan

Winter/Spring 2017

Determining Rate Structure/User Charges

2016 ANNUAL REPORT

CITY OF CADILLAC

ASSET MANAGEMENT

The City has essentially completed the field work for inventory and assessment of sanitary and storm. The Consequence of Failure and Criticality Rating is being derived for each asset and is 90% complete. The capacity for the WWTP, which includes the analysis of the current treatment efficiency, is underway. The Financial Analysis for the rate structure is being performed and is 65% complete. The City is on-track to complete its SAW Grant related asset management activities ahead of the October 2017 SAW deadline.

2017 ANNUAL REPORT
CITY OF CADILLAC
ASSET MANAGEMENT.

City of Cadillac

Wastewater Asset Management Activities for 2018-2019

Report: July 2018

The City of Cadillac is operating and actively maintaining our wastewater collection and treatment system in accordance with our Wastewater Asset Management Plan and Program. Noted below is a summary of our system activities:

Staffing

We maintain the following staff to operate/maintain the wastewater system:

Staff Name	Position	Certification
<i>Jeffrey Dietlin</i>	<i>Director of Utilities/Operator e</i>	<i>Class A</i>
<i>Doug Langworthy</i>	<i>WWTP Assistant Superintendent</i>	<i>Class C</i>
<i>Derek Goodrich</i>	<i>Chief Operator</i>	<i>Class C</i>
<i>Kevin Briggs</i>	<i>Chief Operator</i>	<i>Class C</i>
<i>Chad Marquardt</i>	<i>Operator WWTP</i>	<i>Class C</i>
<i>Dave Skiff</i>	<i>Operator WWTP</i>	<i>Class C</i>
<i>Taysha Laczko</i>	<i>Operator WWTP</i>	<i>Class D</i>
<i>Kevin Moomey</i>	<i>Operator WWTP</i>	<i>N/a</i>
<i>Tim Rossell</i>	<i>Collection System Foreman</i>	<i>D-3, S-1</i>
<i>Pat Bigelow</i>	<i>Collection System Technician</i>	<i>S-3</i>

Inspection/correction activities completed previous year

Project Description	Activity Type	Expenditure
<i>WWTP</i>	<i>Annual O&M</i>	<i>\$879,100</i>
<i>Sewer Cleaning</i>	<i>Annual O&M</i>	<i>\$115,100</i>
<i>Marble Street Sewer</i>	<i>Replacement</i>	<i>\$134,464.33</i>
<i>Garfield Street Sewer</i>	<i>Replacement</i>	<i>\$56,119.29</i>
<i>Blodgett Street Sewer</i>	<i>Replacement</i>	<i>\$20,892.03</i>
<i>Manhole Linings (Cadillac West)</i>	<i>Rehabilitation</i>	<i>\$126,654.30</i>

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Summary of upcoming CIP and O&M activities based on the current
Asset Management Plan

Project Description	Activity Type	Expenditure
WWTP	Annual O&M	\$898,100
Sewer Cleaning	Annual O&M	\$106,400
Cass Street Sewer	Replacement	\$13,234.68
Selma Street Sewer	Replacement	\$84,466.60
Selma/Arthur Street Lift Station	Replacement	\$124,642.42
Miscellaneous Manholes	Rehabilitation	\$25,000

Asset Inventory and Condition Ratings

Condition assessments were made using a variety of methods to develop a Risk of Failure rating for each asset. All ratings were made on a scale of 1-5, with 5 representing the worst condition.

- 3.2 MGD Average Daily Flow Wastewater Treatment Plant. Percentage of treatment plant assets in each Risk of Failure rating category:

1	2	3	4	5
30%	47%	19%	3%	1%

- 11 lift stations. Percentage of lift station assets in each Risk of Failure rating category:

1	2	3	4	5
53%	25%	19%	4%	<1%

- 32,000 feet (6.1 miles) of sanitary force main. The force main was not evaluated during the SAW
- 318,600 feet (60.3 miles) of gravity sanitary sewer. Percentage of gravity sewer pipes in each Risk of Failure rating category:

1	2	3	4	5
58%	6%	7%	8%	21%

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Expenses and Revenues

Total anticipated expenditure for all proposed Operations, Maintenance, and Replacement activities for upcoming year: \$1,889,900

Total annual revenue from current rate structure: \$2,360,000

Excess expenditure to be funded from current Cash Balance: \$300,000

Excess expenditure to be funded through debt issuance: None

Note: Detailed rate analysis including all expenditures/revenues has been compiled by the City's Financial Advisor and is kept on-file at the City Offices.